

ESAI Agent Pipeline

User Guide — How the 10 AI Agents Work Together to Produce Competition Essays

Pipeline Overview

The ESAI suite uses 10 specialized AI agents arranged in a sequential pipeline. Each agent has defined inputs (Needs) and outputs (Produces). The Main Agent orchestrates the flow.

```
Style Profile Builder (once, prerequisite)
    ↓
Main Agent (orchestrates everything below)
    ↓
1. Ideation Agent          → 01_ideation.md
    ↓
2. Research Agent          → 02_research_main.md +
02_research_sources.md
    ↓
3. Parts List Agent        → 02_research_parts_list.md (if hardware)
    ↓
4. Writing Agent           → 03_essay_draft_v{N}.md + outline +
references
    ↓
5. Supervisor Agent        → 07_judge_review_v{N}.md
    ↓ (revision loop back to Writing if needed)
6. Flowchart Agent         → 04_diagrams/
7. Prototype Design Agent → 05_prototypes/
8. UI Design Agent         → 06_uidesign/
    ↓
Ready for submission
```

Agent Details

1. Style Profile Builder

Purpose: Analyzes 5+ winning essay PDFs to produce a reusable writing style profile with executable directives across 8 dimensions.

NEEDS Winning Essay PDFs (user upload)

PRODUCES `00_style_profile.md` — Style Profile

STAGE Foundation — runs once per essay collection, not per competition

Key behavior: Extracts concrete patterns (sentence rhythm, vocabulary fingerprint, opening signatures, evidence integration style). Output is directives, not descriptions.

2. Main Agent (Conductor)

Purpose: Orchestrates the entire pipeline. Inspects state, recommends next skill, supports Interactive and Full Automation modes.

NEEDS Competition Guidebook (optional), Style Profile (optional)

PRODUCES `00_automation_session.md` — Session State

STAGE Meta-conductor — sits above all agents

Key behavior: Two modes: Interactive (user confirms each step) and Full Automation (auto-decides, flags uncertainties, sends Gmail summary). Hard floors: never fabricates citations, stops on disqualifying issues.

3. Ideation Agent

Purpose: Generates 5 rubric-aligned, trend-aware essay angles validated against the guidebook and current news/academic trends.

NEEDS Style Profile + Competition Guidebook

PRODUCES `01_ideation.md` — Ideation Angles with CHOSEN ANGLE

STAGE First substantive skill

Key behavior: Slot strategy (1 safe / 1 distinctive / 1 contrarian / 1 Indonesian-specific / 1 wildcard). Web-searches news + academic sources. Refuses to proceed without guidebook.

4. Research Agent

Purpose: Deep-researches the chosen angle and builds a verified evidence dossier with a gatekept Citation Table.

NEEDS Ideation Output + Style Profile + Guidebook

PRODUCES 02_research_main.md (Research Findings) +
02_research_sources.md (Citation Table)

STAGE Second in pipeline

Key behavior: Every citation requires a verbatim passage + DOI/URL. Minimum 15 sources. A-first quality descent (academic → institutional → news). Counterargument hunting. Rejected candidates logged with reasons.

5. Parts List Agent

Purpose: Builds a verified procurement-ready parts list for physical devices — real prices, dimensions, and sourcing from Indonesian marketplaces.

NEEDS Essay Draft + Guidebook

PRODUCES 02_research_parts_list.md — Parts List

STAGE Optional bridge (hardware essays only)

Key behavior: Indonesian marketplace priority (Tokopedia, Shopee, Bukalapak). Every price/dimension must have a verifiable URL. Subsystem decomposition. Budget tracking. Primary + alternative suppliers.

6. Writing Agent

Purpose: Drafts a polished, rubric-aligned, style-matched essay using only verified citations. Plans image slots for downstream visual agents.

NEEDS Research Findings + Citation Table + Style Profile + Guidebook

PRODUCES 03_essay_draft_v{N}.md — Essay Draft

STAGE Central integration point

Key behavior: 4-checkpoint drafting flow (outline → intro → first body → full draft). Citation gatekeeping (Strict or Batch mode). Image slot markers for Flowchart/Prototype/UI agents. Judge-guided revision mode when supervisor review exists.

7. Supervisor Agent (Judge Panel)

Purpose: Simulates 2–5 judge personas from the rubric. Produces per-criterion scoring, consensus synthesis, score prediction, and prioritized fix list.

NEEDS Essay Draft + Citation Table + Guidebook

PRODUCES 07_judge_review_v{N}.md — Supervisor Review

STAGE Quality-control loop

Key behavior: Quick pass (~5 min) or Deep pass (~20-30 min). Score prediction with placement brackets (competitive / submittable / revise). Go/no-go recommendation. Fix list feeds back into Writing Agent revision mode.

8. Flowchart Agent

Purpose: Generates Mermaid-based diagrams (15 types) with rendered PNG previews, filling image slots in the essay draft.

NEEDS Essay Draft + Research Findings (optional) + Guidebook

PRODUCES .mermaid source + .png renders in 04_diagrams/

STAGE Post-writing visual enrichment

Key behavior: Slot-driven or standalone mode. 15 diagram types (flowchart, sequence, gantt, timeline, pie, mindmap, sankey, etc.). Type-selection logic. Versioned backup before modifying draft.

9. Prototype Design Agent

Purpose: Produces 3D device visualizations using three.js — interactive HTML, static SVG (normal + exploded views).

NEEDS Essay Draft + Research Findings (optional)

PRODUCES `_prototype.html` + `_prototype.svg` in `05_prototypes/`

STAGE Post-writing visual (hardware essays)

Key behavior: Three fidelity tiers (Primitives / Custom geometry / Textured). Parts list integration for real dimensions. Exploded-view toggle. Scene brief required before code generation.

10. UI Design Agent

Purpose: Generates web/mobile/product mockups with a consistent design system — SVGs, interactive HTML, optional 3D device renders.

NEEDS Prototype Design (or Essay Draft)

PRODUCES Device-framed SVGs + Interactive HTML + PNGs in `06_uidesign/`

STAGE Post-writing visual (software/platform essays)

Key behavior: One design system per session (Clean/Institutional/Modern SaaS/Environmental/Academic/Custom). No lorem ipsum — thematic data from essay context. Multi-screen HTML with navigation. 4 slot types: `web_mockup`, `app_mockup`, `product_mockup`, `ui_screen`.

Needs & Produces Summary

Agent	Needs (Inputs)	Produces (Outputs)
Style Profile Builder	Winning Essay PDFs	Style Profile (00_style_profile.md)
Main Agent	Guidebook, Style Profile	Automation Session State
Ideation Agent	Style Profile, Guidebook	Ideation Angles (01_ideation.md)
Research Agent	Ideation Output, Style Profile, Guidebook	Research Findings + Citation Table
Parts List Agent	Essay Draft, Guidebook	Parts List (02_research_parts_list.md)
Writing Agent	Research, Citations, Style Profile, Guidebook	Essay Draft (03_essay_draft_v{N}.md)
Supervisor Agent	Essay Draft, Citations, Guidebook	Judge Review (07_judge_review_v{N}.md)
Flowchart Agent	Essay Draft, Research, Guidebook	Mermaid Diagrams + PNG renders
Prototype Design Agent	Essay Draft, Research	3D HTML + SVG prototypes
UI Design Agent	Prototype/Essay Draft	UI Mockups (SVG + HTML + PNG)

How Users Interact

Starting a Competition

1. Upload competition guidebook (PDF) to the Competition Vault
2. Main Agent checks prerequisites (style profile exists?)
3. Main Agent routes to Ideation Agent
4. User reviews 5 angles, picks one
5. Pipeline continues: Research → Writing → Supervisor → Visuals

The Revision Loop

1. Writing Agent produces draft v1
2. Supervisor Agent reviews → produces fix list + score prediction
3. Writing Agent enters judge-guided revision mode → produces draft v2
4. Supervisor re-reviews → score should improve
5. Once score is in target bracket → proceed to visuals

Visual Enrichment (parallel)

After the draft is stable, three visual agents can run in parallel on their respective image slots:

- **Flowchart Agent** — fills diagram slots (flowchart, timeline, pie, etc.)
- **Prototype Design Agent** — fills prototype slots (3D hardware models)
- **UI Design Agent** — fills web_mockup, app_mockup, product_mockup slots

Full Automation Mode

For overnight runs: Main Agent auto-decides at every checkpoint, flags uncertain decisions with severity tiers ( must review /  should review /  info only), and sends a Gmail summary when done or blocked.